Lab Report: Supervised Learning

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In this lab, we will study the dataset `prostate.data.txt`, which contains nine variables related to a prostate-specific antigen in 97 men about to undergo prostate removal. The objective is to predict the logarithm of this antigen level, represented by the variable *lpsa*, using the eight other variables. We will thus seek to build the best-performing prediction model by comparing several strategies covered in the course, in particular those using linear regression and penalised Ridge and LASSO regressions.

1. Course Reminders and Theoretical Foundations

1.1. Problem formulation

In many practical problems, we study the relationship that may exist between two quantities: $Y \in \mathbb{R}$ (response variable) and $(X_1, \dots, X_p)^T \in \mathbb{R}^p$ (explanatory variables). We then consider a model in which Y is an "affine" function of $(X_1, \dots, X_p)$. In other words:

$$\exists(\beta_0, \beta_1, \dots, \beta_p) \in \mathbb{R}^{p+1} \quad \text{s.t.} \quad Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon, \text{ with } \epsilon \text{ r.v. where } \begin{cases} \mathbb{E}[\epsilon] = 0, \\ \mathbb{V}[\epsilon] = \sigma^2 \end{cases} \tag{1}$$

$$\iff \quad Y = X\beta + \epsilon, \quad \text{with} \begin{cases} X &= (1, X_1, \dots, X_p), \\ \beta &= (\beta_0, \beta_1, \dots, \beta_p)^T \end{cases} \tag{2}$$

Our objective is therefore to find $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$ from the observations $\{(x^i, y^i)\}_{i \in [\![1,n]\!]}$, where for all i, $(x^{i^T}, y^i) \in \mathbb{R}^{p+1} \times \mathbb{R}$. Rewriting the problem, we then wish to solve the following problem with unknown β:

$$Y = X\beta + \epsilon, \quad \text{with} \begin{cases} Y &= (y^1, \dots, y^n)^T \\ X &= \begin{pmatrix} 1 & x_1^1 & \cdots & x_p^1 \\ 1 & x_1^2 & \cdots & x_p^2 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^n & \cdots & x_p^n \end{pmatrix} \\ \beta &= (\beta_0, \dots, \beta_p)^T \\ \epsilon &= (\epsilon^1, \dots, \epsilon^n)^T \quad \text{with the } \epsilon^i \text{ i.i.d. r.v.} \end{cases} \tag{3}$$

1.2. Methods for estimating β

1.2.1. Maximum likelihood

To estimate β by the maximum likelihood method, we must formulate a hypothesis on the joint distribution $p_\beta(x, y)$ of X and Y, before maximising this function with respect to β. However, we adopt a statistical learning approach where we will make no assumption on the joint distribution! Consequently, we cannot use the maximum likelihood method.

1.2.2. Statistical learning method with the least squares estimator

We can therefore estimate β with the least squares estimator $\hat{\beta}^{MC}$:

$$\hat{\beta}^{MC} = \text{argmin}_{\beta \in \mathbb{R}^{p+1}} \frac{1}{2} \sum_{i=1}^{n} (y^i - x^i \beta)^2 \quad \forall i \in [\![1, n]\!],\ y^i - x^i\beta \text{ is called the } i^{\text{th}} \text{ residual.} \tag{4}$$

$$= \text{argmin}_{\beta \in \mathbb{R}^{p+1}} \frac{1}{2} \|Y - X\beta\|_2^2 \tag{5}$$

Expanding the expression we wish to minimise using the binomial identity:

$$\frac{1}{2}\|Y - X\beta\|_2^2 = \frac{1}{2}\left(\|Y\|_2^2 + \|X\beta\|_2^2 - 2\langle Y, X\beta\rangle\right) \tag{6}$$

$$= \frac{1}{2}\left(\|Y\|_2^2 + (X\beta)^T(X\beta) - 2\langle Y, X\beta\rangle\right) \tag{7}$$

$$= \frac{1}{2}\left(\|Y\|_2^2 + \beta^T X^T X\beta - 2Y^T X\beta\right) \tag{8}$$

$\beta \longmapsto \frac{1}{2}\left(\|Y\|_2^2 + \beta^T X^T X\beta - 2Y^T X\beta\right)$ is differentiable on $\mathbb{R}^{p+1}$. The function attains its minimum value at $\hat{\beta}^{MC}$ if and only if:

$$\nabla\left(\frac{1}{2}\|Y - X\beta\|_2^2\right)\Big|_{\beta=\hat{\beta}^{MC}} = \frac{1}{2}\left(2X^T X\hat{\beta}^{MC} - 2X^T Y\right) = X^T X\hat{\beta}^{MC} - X^T Y = 0 \tag{9}$$

$$\iff \boxed{\beta^{MC} = (X^T X)^{-1} X^T Y \quad \text{if } X^T X \text{ is invertible. This is the least squares estimator.}} \tag{10}$$

Since X and β are deterministic, we have $\mathbb{E}[Y] = \mathbb{E}[X\beta + \epsilon] = X\beta$ because ϵ is centred. Moreover, $\mathbb{V}[Y] = \sigma^2 I$ since $\mathbb{V}[X] = \sigma^2 I$. It then follows:

$$\mathbb{E}[\hat{\beta}^{MC}] = \mathbb{E}[(X^T X)^{-1} X^T Y] = (X^T X)^{-1} X^T \mathbb{E}[Y] = (X^T X)^{-1} X^T X\beta = \beta, \tag{11}$$

$$\mathbb{V}[\hat{\beta}^{MC}] = \mathbb{V}[(X^T X)^{-1} X^T Y] = (X^T X)^{-1} X^T \mathbb{V}[Y] X (X^T X)^{-1} = \sigma^2 (X^T X)^{-1} \tag{12}$$

The least squares estimator is a BLUE (Best Linear Unbiased Estimator). If ϵ follows a normal distribution, then this estimator is efficient, meaning that among all estimators of β, it is the one whose variance attains the Cramér-Rao bound.

1.2.3. Statistical learning method with penalised Ridge and LASSO regression when $X^T X$ is not invertible

If $X^T X$ is not invertible, we can use Ridge and LASSO regression methods. Here are the estimators for each of the two methods:

$$\hat{\beta}^{\text{Ridge}} = \text{argmin}_{\beta \in \mathbb{R}^{p+1}} \left(\|Y - X\beta\|^2 + \lambda\|\beta\|_2^2\right), \quad \text{with } \lambda > 0 \tag{13}$$

Analogously to section 1.2.2., we find the explicit expression for the estimator:

$$\iff \boxed{\beta^{\text{Ridge}} = (X^T X + \lambda I_{p+1})^{-1} X^T Y, \quad \text{with } \lambda > 0} \tag{14}$$

Comparing with the expression in (10), we note that we have slightly perturbed $X^T X$ so as to make it invertible. As for the LASSO estimator:

$$\hat{\beta}^{\text{LASSO}} = \text{argmin}_{\beta \in \mathbb{R}^{p+1}} \left(\|Y - X\beta\|^2 + \lambda\|\beta\|_1\right), \quad \text{with } \lambda > 0 \tag{15}$$

For the LASSO estimator, there is no explicit expression; it is generally approximated using numerical methods. <u>These estimators introduce a bias, but reduce the variance, which decreases overfitting</u>!

2. Study of `prostate.data.txt`

2.1. Presentation of the dataset

This study concerns a dataset called `prostate.data.txt`, which contains information on 97 individuals about to undergo prostate removal. Each individual is described by 8 explanatory variables and one response variable *lpsa*. The dataset also includes an unnamed column corresponding to the individual identifier, as well as a *train* column that allows the dataset to be split into training and test sets.

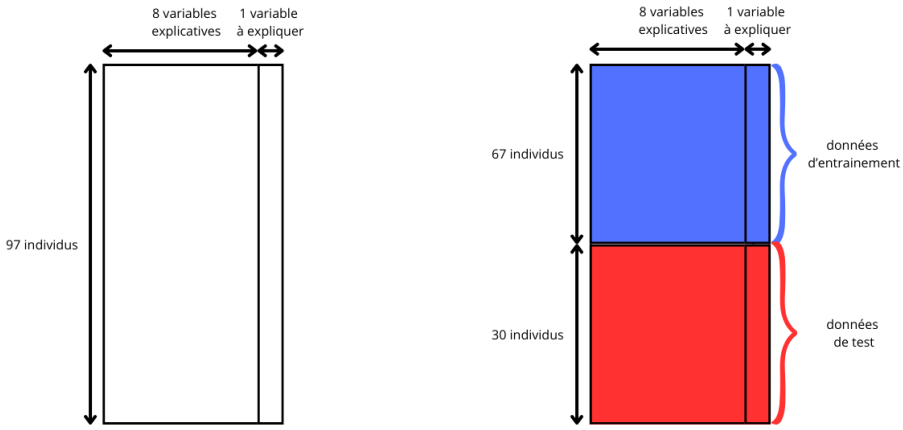

Figure 1: Data pre-processing. We remove the unnamed column before splitting the dataset into training and test data according to the *train* column, which we also remove.

Variable	Description
lcavol	Logarithm of cancer volume
lweight	Logarithm of prostate weight
age	Patient age
lbph	Logarithmic measure related to benign prostatic hyperplasia
svi	Seminal vesicle invasion indicator, 0 or 1
lcp	Another logarithmic measure related to cancer
gleason	Gleason score, cancer aggressiveness
pgg45	Percentage of Gleason grade
lpsa	Logarithm of PSA

Table 1: Probable interpretation of the nine quantitative variables of the `prostate.data.txt` dataset.

2.2. Methodology for building the best prediction model for *lpsa*

2.2.1. Forward and backward variable selection

As suggested in the problem statement, we will perform variable selection. Indeed, in many data problems, we work with high-dimensional datasets, and not all explanatory variables are equally important, as we potentially have more data than necessary. As a result, when training prediction models, we cannot afford to train the model on all explanatory variables, as the computational costs would be too high to account for superfluous variables that do not add significant value. To select these variables, there are two methods: forward and backward variable selection.

Forward variable selection Forward variable selection consists in choosing a criterion to evaluate the importance of explanatory variables, taking for example the BIC. We start with an empty dataset and add variables iteratively. First, since we have not yet chosen any penalisation coefficients, we train 8 linear regression models, each using a different explanatory variable. We then compare these models to select the variable that minimises the BIC. Once the first variable is chosen, we add a

second variable from the remaining 7 and retrain 7 new linear regression models. We select the second variable such that the pair formed with the first optimises the BIC among the 7 options, while ensuring that the BIC obtained is lower than that of the single-variable model. We continue this process until adding a new variable no longer decreases the BIC.

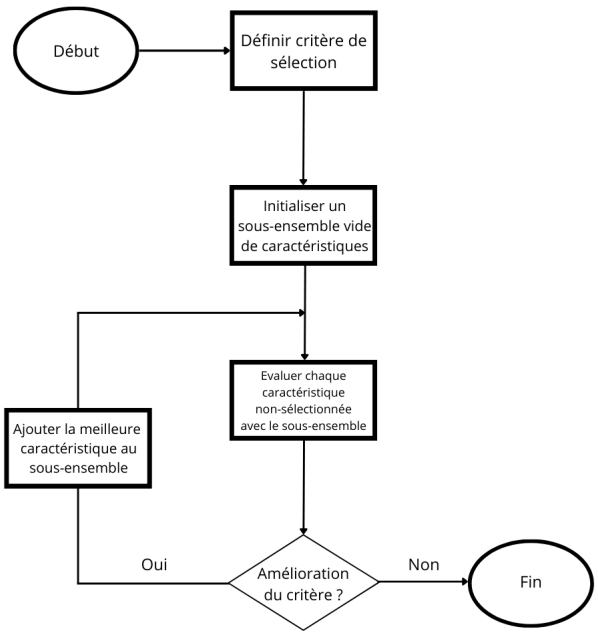

Figure 2: Forward variable selection method, also known as greedy feature selection.

Backward variable selection (1/2) Backward variable selection consists in choosing a criterion to evaluate the importance of explanatory variables, taking for example the BIC. We start with a complete dataset including all explanatory variables. First, we train a linear regression model with all the variables. Then, we iteratively remove one variable at a time, evaluating the impact on the BIC. We begin by identifying the variable whose removal increases the BIC the least / decreases it the most. Once this first variable is eliminated, we retrain 7 linear regression models with the remaining variables and repeat the process. At each step, we verify that removing a variable does not significantly increase the BIC, seeking to maintain the model's performance. We continue this progressive elimination until no further removal can improve the selection criterion.

Backward variable selection (2/2) Since we are training a linear regression model on all variables anyway, we can also examine the coefficients of $\hat{\beta}^{MC}$. Indeed, if the coefficients are centred around zero, this may indicate low importance of the variable, whereas a coefficient far from zero signals a significant, positive or negative, contribution of the variable. It is also important to note that greedy selection methods, whether forward or backward, offer a local optimisation of the BIC or AIC and do not necessarily guarantee a global optimum.

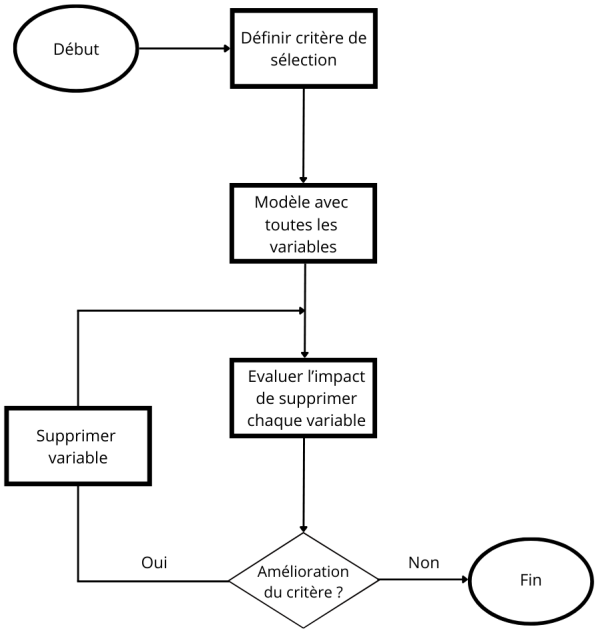

Figure 3: Backward variable selection method (1/2).

Cross-validation for computing BIC and AIC The BIC and AIC criteria require the computation of the likelihood. In practice, they are traditionally computed by training a model and evaluating the mean squared error between the training target and the predicted target, which provides a BIC and AIC value. However, these initial BIC and AIC values are arbitrary as they are computed on the entire training set. To reduce this bias and improve variable selection, we propose an alternative method: using cross-validation with $K = 3$ folds. This approach consists in arbitrarily dividing the dataset into 3 subsets, then computing the BIC and AIC on each fold before averaging them. This method yields a more robust estimate that is less dependent on a single sample, thereby reducing the risk of overfitting.

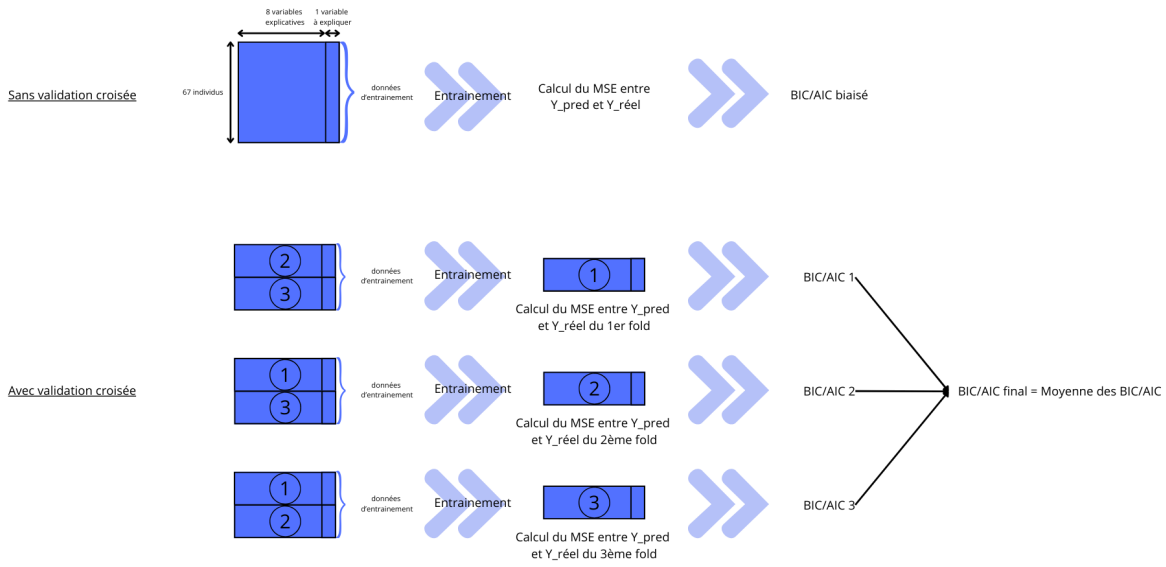

Figure 4: Comparison of BIC and AIC computation with and without cross-validation.

2.2.2. How to choose λ to correctly penalise the regressions?

Once the variables are selected, we must determine the penalisation coefficients λ for the penalised Ridge and LASSO regressions. To do so, we use 3-fold cross-validation as before. This choice avoids obtaining biased λ values that could overfit the entire training set and thus compromise the model's generalisation. At each iteration, we train penalised regression models on the first two folds by testing a wide range of λ values, generating many models. We then evaluate the performance of these models on the third fold, identifying the one that minimises our error criterion. The λ associated with this model is then considered optimal for that iteration. We repeat this process across all three folds and take the average of the obtained λ values to define our final penalisation coefficient. Ultimately, this method allows us to select six different coefficients, hence six different models: three for Ridge regression and three for LASSO regression, depending on the error criterion to optimise, whether MSE, MAE, or R^2.

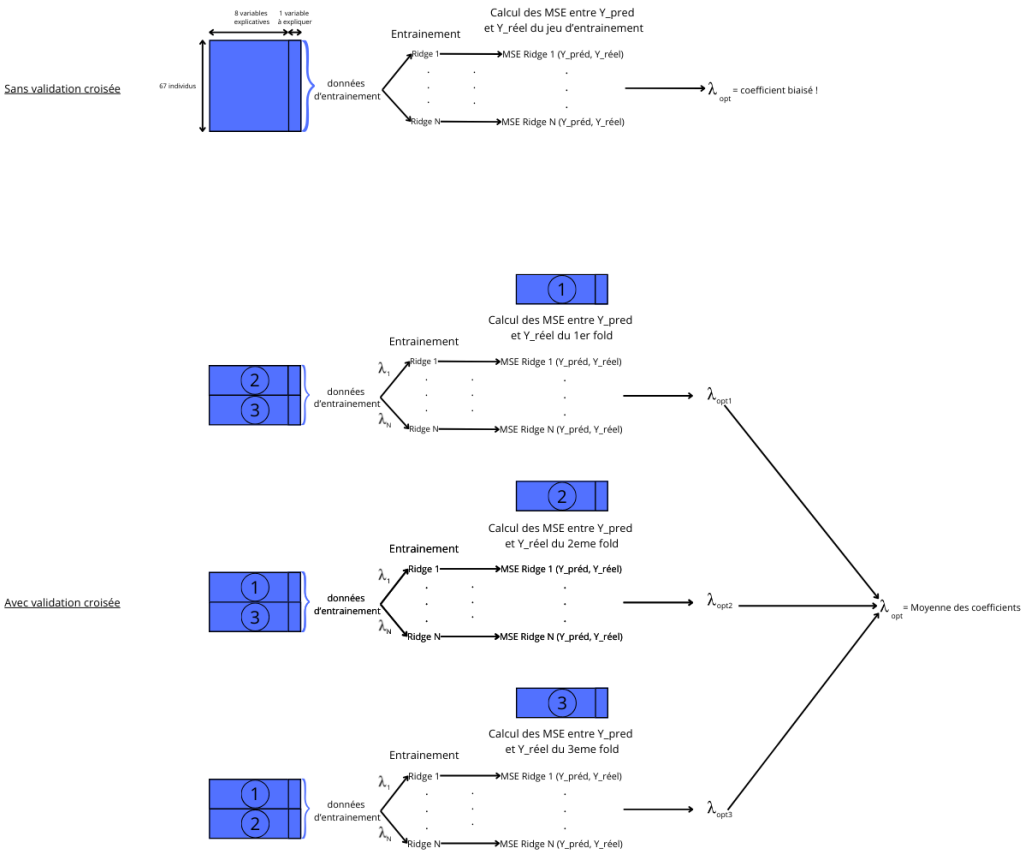

Figure 5: Comparison of determining the best penalisation coefficient λ with and without cross-validation. Here, example of Ridge for MSE.

2.2.3. Evaluating the performance of the studied models

Now that we have selected the explanatory variables and determined the penalisation coefficients, i.e. our hyperparameters, we evaluate the final performance of each model to choose the best-performing one. To do so, we train the six models one last time on the full training set, without cross-validation. We then apply each model to the test set to predict the variable *lpsa*. Comparing the obtained predictions, `ypred`, with the actual values, `ytest`, will allow us to quantify the error of each model and evaluate their final performance. The model with the lowest error will be retained as the optimal solution for our problem.

Ideally, the split into training and test sets of the `prostate.data.txt` dataset should also have

been subjected to cross-validation, allowing the performance results to be averaged. However, we will follow the instructions of the problem statement, especially since the current training/test split appears visually homogeneous.

3. Results

3.1. The explanatory variables

The greedy variable selection methods, forward and backward, according to the BIC and AIC, described in section 2.2.1, indicate that we should retain the two explanatory variables *lcavol* and *lweight*. This result was obtained by running the code several times, thus modifying the folds at each iteration.

Furthermore, we also performed a linear regression using all available data and examined the coefficients of $\hat{\beta}^{MC}$. The coefficient values associated with the variables *lcavol* and *lweight* are indeed higher in absolute value than those of the other coefficients, except for the coefficient associated with the variable *svi*. We nevertheless choose not to include *svi* in our new training set, in order to adhere to the methodology.

Indeed, greedy variable selection methods only optimise the BIC and AIC locally, which may lead to a different global optimum. Thus, a trade-off is established between prediction accuracy and execution speed.

3.2. The penalisation coefficients

We performed 3-fold cross-validation for λ values between 10^{-4} and 10^{2}. The optimal λ values obtained are as follows:

λ	MSE	MAE	R^2
Ridge regression	1.778966	10.971005	9.493520
LASSO regression	1.345739	8.136306	5.432569

Table 2: Summary table of penalisation coefficients.

At each iteration of the 3-fold, a large variability in the coefficients appears to arise; this could indicate that the amount of data is too small. Indeed, the optimisation criteria are computed on a fold of approximately 20 data points, which is below the quantities typically used.

λ	Ridge MSE	Ridge MAE	Ridge R^2
$\hat{\beta}^{\text{Ridge}}$	$\begin{pmatrix} 0.637624 \\ 0.44921489 \end{pmatrix}$	$\begin{pmatrix} 0.59721869 \\ 0.45940182 \end{pmatrix}$	$\begin{pmatrix} 0.60319754 \\ 0.45796681 \end{pmatrix}$

Table 3: Summary table of $\hat{\beta}^{\text{Ridge}}$ values.

λ	LASSO MSE	LASSO MAE	LASSO R^2
$\hat{\beta}^{\text{LASSO}}$	$\begin{pmatrix} 0.63736502 \\ 0.44826275 \end{pmatrix}$	$\begin{pmatrix} 0.59275981 \\ 0.45807281 \end{pmatrix}$	$\begin{pmatrix} 0.61064978 \\ 0.45448688 \end{pmatrix}$

Table 4: Summary table of $\hat{\beta}^{\text{LASSO}}$ values.

The coordinate values show that the coefficients are not zero, indicating that we have successfully captured significant relationships in our model. However, it is possible that some relevant explanatory variables have been omitted. This choice is intentional: with greedy variable selection, we accept sacrificing a little precision in order to reduce algorithm execution time.

We also performed the penalised regressions with these coefficients on the entire dataset. We observed that both methods favour coefficient sparsity, but in different ways: the Ridge model applies a penalty on the sum of squared coefficients, which tends to reduce the coefficient values without setting them to zero entirely. This allows all variables to be retained while attenuating the impact of the less relevant ones. In contrast, the LASSO model applies a penalty on the sum of absolute values of the coefficients, which pushes some of them to zero. This approach therefore induces stricter variable selection, retaining only those that make a significant contribution. Thus, LASSO generally leads to a more parsimonious model, while Ridge favours smooth regularisation without complete variable elimination.

3.3. Model performance

Error	MSE	MAE	R^2
Standard linear regression	0.478136	0.519630	0.544478
Ridge regression	0.479413	0.517812	0.536754
LASSO regression	0.479383	0.517062	0.538346

Table 5: Summary table of mean squared errors, mean absolute errors and coefficients of determination R^2 for the six studied models and the linear regression model.

The simple linear estimator yields the best results in terms of MSE and R^2, while the LASSO estimator with $\lambda = 8.136306$ achieves the best results in MAE. As mentioned in section 1.2.3, the penalised estimators introduce bias to reduce overfitting and improve model robustness, particularly when there are less relevant features or high variance.

However, here the performance difference is minimal between the penalised estimators and the standard linear model, suggesting that overfitting was not a major issue in our dataset. This could indicate either that the data is already well suited to a linear model, or that the number of variables is not high enough to cause significant overfitting. Consequently, adding penalties does not provide a substantial benefit in this situation.

Conclusion

In conclusion, this study on the `prostate.data.txt` data reveals several significant methodological limitations. The large variance observed in the λ coefficients can be attributed to the small volume of available data, which makes the results unstable. Nevertheless, we observe consistency in the selection of explanatory variables across the different methods.

The fixed split between training and test data, without cross-validation at this level, constitutes an important weakness, although the split appears homogeneous at first glance.

Our approach remains highly experimental, with limited theoretical support. The absence of clearly differentiated results between the methods is therefore not surprising, reflecting the limitations of our approach and the inherent complexity of analysing small datasets.